

OWENS CORNING®

Thermafiber® FireSpan® 90 & FireSpan® 40

Mineral Wool Insulation – Curtain Wall & Perimeter Fire Containment

Product Description

Thermafiber® FireSpan® products are specifically engineered to provide enhanced life safety fire protection in perimeter fire containment systems and curtain wall assemblies. These non-combustible, moisture-resistant, non-corrosive, and mildew-resistant mineral wool insulation panels are designed to perform under extreme conditions.

Their natural dark color provides essential shadowing behind opaque glass panels (spandrels) where they are installed, preventing visual read-through. FireSpan® 40 and 90 provide exceptional thermal insulation, acoustical performance, and vapor control in various 1, 2, and 3-hour fire-rated systems listed by UL and Intertek.

Technical Data & Physical Properties

The following table summarizes the key performance specifications for FireSpan® 40 and FireSpan® 90, tested under strict standard procedures.

Product	Nomk-Value at 75°F(24°C) BTU [†]	i-nal Den-sity	R-Valueper inch	Unfaced (ASTM E84)		Foil-Faced (ASTM E84)	
				Flame	Smoke	Flame	Smoke
FireSpan 40	4.0	0.23	4.3	0	0	25	0
	pcf						
	(64 kg/m³)						
FireSpan 90	8.0	0.23	4.3	0	0	25	0
	pcf						
	(128 kg/m³)						

[†] Thermal Conductivity (k) measured in BTU·in / hr·sq.ft·°F. R-Value is thickness divided by 'k'.

Features & Product Benefits

- **Exceptional Fire Resistance:** Rated for temperatures exceeding 2,000°F (1,093°C).
- **Energy Efficiency:** Helps lower thermal transfer to save building energy and reduce greenhouse gas emissions.
- **Acoustical Performance:** Significantly dampens sound transmission across perimeter wall cavities.
- **Moisture & Mold Control:** Non-corrosive, moisture-resistant, and inherently resistant to mold/termite growth.
- **Sustainable Design:** Contains a minimum of 70% recycled content (certified by ICC-ES VAR-1025), contributing directly to LEED® and Green Globes® rating points.

Standards & Code Compliance

Thermafiber® FireSpan® complies with a wide range of building standards and regulatory requirements:

- **ASTM C 665:** Type I, III (Class A, Category 1)
- **ASTM C 612:** FireSpan 40 & FireSpan 90: Type IA, IB, II, III, IVA
- **ASTM E 136 / CAN/ULC S114:** Rated non-combustible as defined by NFPA Standard 220
- **ASTM E 96:** Unfaced tested at 50 Perms; Foil-faced tested at 0.02 Perms (acting as a Class I vapor retarder)
- **ASTM C 1104:** Moisture adsorption/absorption less than 1% by volume

- **ASTM C 356:** Linear shrinkage less than 2% at 650°C (1200°F)
- **ASTM E 2307:** Approved for use in listed perimeter fire containment systems (Aluminum, Drywall, Glass, Granite, and Precast Concrete Spandrel designs)

Product Availability

Product	Available Thicknesses	Standard Widths	Standard Lengths
FireSpan 40	2" – 7" (in 1/2" increments)	24", 36", 72"	48", 60", 72"
FireSpan 90	1" – 7" (in 1/2" increments)	24", 36", 72"	48", 60", 72"
Tolerances	+1/4", –1/8"	±1/8"	+3/4", –1/4"

Installation Guidelines

FireSpan® insulation panels must be mechanically secured to horizontal and vertical aluminum mullions using architect-approved mechanical insulation fasteners. Horizontal stiffeners (such as T-bars, L-angles, or channel sections) must secure the FireSpan® panels directly behind the spandrel area at the floor slab level.

For fast and efficient assembly, the patented **Thermafiber® Impasse® System** hanger hardware can be utilized to simplify panel mounting in modern curtain wall frameworks.